

[Final Year Project Proposal]

Sr#	Student Name	Roll Number	Credit Completed	Signature
1	Seemab Anas	22P-9210	100	
2	Muhammad Hammad	22P-9206	100	
3	Muhammad Hussain	22P-9042	94	

Suggested Supervisor:Faculty Member's Name: Shoaib Muhammad Khan

Signature: _____

Date (09 September 2025)

Project Details

Project Title	AI-Powered Clinical Assistant for Real-Time Documentation and Decision Support		
Project Area of Specialization	Artificial Intelligence, Natural Language Processing, and Healthcare Informatics		
List Related Core Subjects	1) Artificial Intelligence 2) Software Design and Analysis 3) Database Systems 4) Software Engineering		
List Related Elective Subjects	1) Natural Language Processing 2) Data Science		
Project Start Date	2025-08-10	Project End Date	2026-05-31
Project Summary (less than 2500 characters)	Doctors spend over 60% of consultation time on documentation, which reduces efficiency and increases the risk of missed details, incomplete safety checks, and diagnostic errors. This administrative burden leads to cognitive overload, inconsistent clinical documentation, and reduced focus on patient care. The proposed AI-Powered Clinical Assistant is an innovative healthcare technology solution that works alongside existing hospital systems to reduce documentation burden and strengthen clinical decision-making. The system performs real-time medical transcription with high accuracy for medical terminology, automatically extracts key clinical entities such as symptoms, allergies, medications, and past conditions, and generates structured clinical notes. Beyond documentation automation, the system guidelines andn intelligent safety layer by flagging potential drug interactions,		

	highlighting missing vital information, suggesting differential diagnoses based on evidence-based clinical guidelines, and providing treatment recommendations with confidence scores. The AI assistant includes advanced medication intelligence with multi-level drug interaction analysis, contraindication checking, and alternative medication suggestions. Built using cutting-edge technologies the system aims to reduce documentation time by 30-40% while improving diagnostic accuracy and ensuring safer patient care. This project demonstrates practical application of artificial intelligence and natural language processing in solving real-world healthcare challenges without disrupting existing clinical workflows.
Project Objectives (less than 2500 characters)	1. Develop a Real-Time Clinical Documentation Assistant - Implement medical speech-to-text transcription with higher accuracy for medical terminology - Extract critical clinical entities such as symptoms, allergies, medications, past conditions, and vital signs - Generate structured clinical notes automatically - Create intelligent session summarization with key findings and action items 2. Enhance Diagnostic Accuracy and Clinical Safety - Integrate an AI-powered diagnostic safety net to cross-check symptoms against comprehensive medical databases - Provide differential diagnosis suggestions with confidence scores and severity assessments - Flag potential conditions that might be overlooked during consultations - Ensure patient safety by highlighting missing vital information and documentation inconsistencies - Implement evidence-based clinical decision support aligned with medical guidelines 3. Implement Advanced Medication Intelligence Engine - Perform comprehensive drug-drug, drug-food, and drug-condition interaction checks in real-time - Suggest safe alternative medications and optimize dosages based on patient-specific factors. - Integrate intelligent alerts for contraindications and allergy risks with visual indicators - Provide cost-effectiveness analysis for treatment options and insurance formulary checking

	4. Provide Comprehensive Clinical Decision Support Recommend evidence-based treatments and interventions following established clinical protocols Suggest appropriate laboratory tests, specialist referrals, and follow-up appointments when indicated Support healthcare providers with explainable AI recommendations and confidence scoring Implement risk stratification algorithms for patient condition assessment 5. Evaluate Clinical and Technical Performance Impact Measure quantifiable improvements in documentation efficiency, accuracy, and clinical workflow optimization Benchmark comprehensive system performance including speech recognition accuracy, response latency, and system uptime Conduct thorough validation testing and collect detailed feedback from healthcare professionals to assess usability and clinical effectiveness
Project Implementation Method (less than 2500 characters)	Core System Development Implement high-accuracy real-time speech-to-text conversion specifically optimized for medical vocabulary and terminology Develop comprehensive natural language processing (NLP) pipeline to identify and extract critical clinical information including symptoms, medical history, medications, and patient concerns Create automated structured consultation summary generation system to significantly reduce doctors' documentation workload Build intuitive user interface allowing healthcare providers to view, verify, and edit generated clinical notes during or immediately after patient consultations. Intelligence and Safety Features Enhance NLP models for superior accuracy in complex medical terminology recognition and clinical context understanding Integrate sophisticated diagnostic support component that cross-references patient symptoms with comprehensive medical knowledge sources to highlight possible conditions and differential diagnoses Implement advanced medication safety engine performing multi-level checks for drug interactions, contraindications, and allergy risks with real-time alerting Develop clinical decision support system providing evidence-

	based recommendations and alerts without disrupting established clinical workflows Integration and Optimization Develop comprehensive interoperability modules with compliant APIs enabling seamless integration with existing Electronic Health Record (EHR) systems and healthcare platforms Introducing advanced analytics dashboard providing system monitoring, usage insights, performance metrics, and clinical outcome tracking Strengthen data privacy and security mechanisms ensuring full compliance with healthcare regulations standards Optimize system architecture for minimal latency, high availability, and horizontal scalability to handle real-time clinical environment demands Conduct comprehensive pilot testing and validation using synthetic medical data and controlled case studies to assess clinical effectiveness, system reliability, and user satisfaction
Benefits of the Project (less than 2500 characters)	1. Improved Efficiency in Clinical Workflows Automates comprehensive note-taking and clinical documentation, reducing administrative time doctors spend on routine tasks by 30-40% Enables significantly faster patient consultations by streamlining information capture, organization, and retrieval processes Minimizes duplication of effort by generating structured, reusable summaries compatible across multiple healthcare systems and platforms Standardizes clinical documentation following established medical formats ensuring consistency and compliance 2. Enhanced Diagnostic Accuracy and Patient Safety Assists healthcare providers by intelligently highlighting overlooked symptoms, missing critical information, or potential diagnostic considerations during patient consultations Provides robust safety layer through comprehensive medication interaction checks, contraindication alerts, and allergy risk assessments Supports evidence-based clinical decision-making with AI-driven diagnostic suggestions backed by medical literature and clinical guidelines Reduces medical errors through systematic validation and cross-referencing of clinical information

	3. Reduced Cognitive Burden on Healthcare Providers Significantly decreases mental workload by automatically organizing, structuring, and presenting relevant patient data in digestible formats Offers intelligent real-time recommendations and clinical insights, allowing doctors to focus on critical thinking and patient interaction Lowering the risk of human error and information omission and increasing time available for direct patient care 4. Seamless Integration into Existing Healthcare Ecosystems Designed specifically to complement and enhance current Electronic Health Record (EHR) systems rather than requiring disruptive replacements API-driven architecture ensures broad adaptability and compatibility with multiple healthcare platforms and vendor systems
Technical Details of Final Deliverable (less than 2500 characters)	Core System Functionalities Advanced Speech-to-Text Processing: Real-time transcription of doctor–patient consultations with specialized optimization for complex medical terminology, abbreviations, and clinical language patterns Comprehensive NLP Engine: Sophisticated extraction of critical clinical entities including symptoms, diagnoses, allergies, medications, family history, and social factors from free-text conversations Intelligent Automated Summarization: Generation of structured consultation summaries following standards to dramatically reduce documentation time and improve consistency Advanced Medication Intelligence Module: Multi-level drug interaction analysis, comprehensive allergy alerts, contraindication detection, alternative medication suggestions, and dosage optimization based on patient-specific factors Evidence-Based Clinical Decision Support: AI-driven differential diagnosis suggestions, treatment recommendations, specialist referral guidance, and intelligent reminders for missing or incomplete clinical data Robust System Architecture Microservices Design: Independent, scalable components for transcription, NLP processing, safety checks, and clinical recommendations ensuring system flexibility, maintainability, and

	fault tolerance Healthcare Interoperability: API-first architecture enabling seamless integration with Electronic Health Record (EHR) systems and other clinical platforms Multi-Platform User Interface: Responsive web accessible dashboards displaying consultation summaries, intelligent alerts, and clinical recommendations in intuitive, user-friendly formats optimized for clinical workflows Comprehensive Analytics Layer: Advanced performance tracking, system accuracy monitoring, usage analytics, and clinical outcome insights for healthcare providers and system administrators Security and Regulatory Compliance Data Protection: Encryption for all patient communications, secure data transmission protocols, and encrypted database storage Access Control: Role-based authentication system with granular permissions for authorized healthcare personnel Regulatory Compliance: Full adherence to healthcare data privacy standards fulfilling relevant regulatory requirements Expected Technical Deliverable A production-ready AI Clinical Assistant prototype capable of accurate real-time transcription and clinical information extraction, providing meaningful safety alerts and evidence-based recommendations during patient consultations, seamless integration with external healthcare systems through standardized APIs, and reliable, responsive interface optimized for clinical environments.
Final Deliverable of the Project	1. AI Clinical Assistant Prototype – Fully functional software system with real-time transcription, medical information extraction, and intelligent consultation summarization 2. Advanced Medication Safety Module – Comprehensive automated checking of drug interactions, contraindications, allergy alerts, and alternative medication recommendations 3. Clinical Decision Support System – Evidence-based diagnostic suggestions, treatment recommendations, and intelligent safety reminders based on extracted clinical data 4. Healthcare Integration Layer –Compliant API modules enabling seamless connectivity with Electronic Health Record (EHR) systems and external healthcare platforms 5. Analytics & Performance Monitoring – Comprehensive dashboards for system performance tracking, clinical accuracy

	assessment, and usage analytics 6. Complete Documentation Package – Technical documentation, user manuals, API documentation, deployment guides, and system administration instructions 7. Comprehensive Evaluation Report – Performance benchmarks, clinical validation results, user feedback analysis, and recommendations for future enhancements
Type of Industry	Healthcare and Medical Technology (HealthTech) Artificial Intelligence and Data Science in Healthcare, Clinical Decision Support Systems
Technologies	Frontend Development: Next.js with TypeScript and Tailwind CSS for building a responsive web interface and clinical dashboard. Backend Development: Node.js with Next.js API routes connected to a PostgreSQL database (via Prisma ORM) for structured clinical data management. AI & NLP: OpenAI Whisper API for speech-to-text transcription and GPT models (3.5/4) for text analysis, summarization, and decision support. Hugging Face Transformers and spaCy for medical entity extraction. Real-Time Communication: Socket.io for live transcription streaming and instant clinical alerts. Medical Data Integration: FHIR R4 standards for healthcare interoperability and RxNorm API for medication data and interaction checking. Authentication & Security: NextAuth.js for authentication, role-based access control, encryption for data protection. Testing & Development: Jest for testing, Prisma for database management, and Swagger/OpenAPI for documenting APIs. Database & Storage: PostgreSQL for structured data storage, Redis for caching and session handling, and local storage for audio files. Monitoring & Analytics: Winston logging and a simple analytics dashboard for system usage and performance tracking. Additional Libraries: React Hook Form for form validation, Lodash for utility functions, and Recharts for data visualization.

Sustainable Development Goals	SDG 3: Good Health and Well-Being Contributes to improved healthcare delivery by reducing medical errors, enhancing diagnostic accuracy, and enabling safer clinical decision-making Helps healthcare providers spend less time on administrative documentation and more focused time on direct patient care, ultimately improving overall health outcomes and patient satisfaction Supports healthcare accessibility by streamlining clinical workflows and reducing barriers to efficient medical care SDG 9: Industry, Innovation, and Infrastructure Promotes significant innovation in clinical practice by integrating cutting-edge artificial intelligence with existing healthcare systems Creates scalable and interoperable digital infrastructure supporting healthcare digital transformation and establishing foundation for future intelligent health technologies Advances research and development in medical AI applications contributing to global healthcare technology advancement
--------------------------------------	--

Project Key Milestones

Elapsed time in (days or weeks or month or quarter) since start of the project	Milestone	Deliverable
Month 1 - 2	Requirements Analysis & System Design	Project plan, literature review, technical specifications, system architecture diagrams
Month 3 - 4	Core Backend & NLP Pipeline Development	Database schema, authentication system, basic API endpoints, real-time transcription, medical entity recognition module
Month 5 – 6	Frontend Interface & Note Generation	Web application with transcription display, automated note generation, and backend integration
Month 7 – 8	AI Diagnostic & Medication Intelligence Features	Symptom analysis engine, diagnostic suggestion module with confidence scoring, and medication safety features
Month 9 – 10	System Integration & Advanced Features	FHIR API implementation for healthcare interoperability, enhanced analytics dashboard, and system-level validation framework
Month 11 - 12	Final Testing, Optimization & Deployment	Comprehensive testing suite, performance optimization, analytics dashboard, final documentation, deployment package, and evaluation report

Project Equipment Details

Item Name	Type	No. of Units	Per Unit Cost (in Rs)	Total (in Rs)
OpenAI Whisper API (Speech-to-Text)	API Subscription	1 (approx. 60 hrs transcription)	108 approx. per hr	6500
OpenAI GPT-3.5/4 API Credits	API Subscription	1 (approx. 5M tokens for 1 year)	12000	12000
RxNorm / Medical Data APIs	API Subscription	1	8000	8000
PostgreSQL Database (Managed Service)	Software/Hosting	1 (12 months)	1000	12000

Redis (Caching & Session Management)	Software/Hosting	1 (12 months)	500	6000
Development Tools & Libraries (e.g., IDE, testing frameworks, logging, visualization tools)	Software	1	10000	10000
Miscellaneous Costs (extra API usage, data storage)	Miscellaneous	1	8000	8000

Total: 62,500 PKR ONLY